Gholamali Aminian

Team Lead, Turing

Address: London, UK
Email: gholamali.aminian@gmail.com
Phone: +44 7920045415
Google Scholar: Gholamali Aminian Scholar
Website: https://gholamaliaminian.github.io

Objective

As a **Team Lead** at **Turing**, I currently work on RSI benchmarks. Before that, I led the team responsible for quality control of LLM-based agentic tasks delivered to frontier AI labs: designing QC pipelines and adversarial, agentic-based evaluation methods, detecting reward hacking by agents, identifying ambiguity in task instructions, detecting misalignment between instructions and verifiers, assessing test coverage, and designing evaluations that reveal meaningful failure modes rather than merely confirm success on easy cases. I also work on benchmarking, hill-climbing on benchmarks and evaluating models and agents, most recently ML4Science-Bench, a benchmark of end-to-end machine-learning tasks for science (NeurIPS 2026 AI for Science Workshop, Oral).

My research background is central to this work. I bring a decade of experience in information theory, **generalization**, and **alignment**, including research on KL-regularized RLHF and best-of-n methods (NeurIPS 2025; ICLR 2026), as well as **off-policy evaluation** (ICML 2025 Spotlight). I have applied the same scientific discipline in practice as an **AI Data Scientist at HSBC**, where I developed FraudTransformer (CAI 2026), and as a Research Associate at The Alan Turing Institute.

I am interested in roles involving LLM benchmarking and agentic AI, where rigorous evaluation, alignment, and inference-time control are essential to understanding model capabilities and building trustworthy AI systems.

Employment

Team Lead — June 2026 – Present
Turing, US (Remote)

Research Associate — July 2022 – June 2026
The Alan Turing Institute, London, UK

AI Data Scientist — Nov 2024 – Aug 2025
HSBC, London, UK

Honorary Research Postdoctoral Fellow — March 2020 – July 2022
University College London (UCL)
Newton International Fellowship

Research Assistant — September 2017 – March 2020
Sharif University of Technology

Education

Ph.D. in Communication Systems — 2012 – 2017
Sharif University of Technology
GPA: 18.37/20

M.Sc. in Communication Systems — 2010 – 2012
Sharif University of Technology
GPA: 17.76/20

B.Sc. in Electrical Engineering 2006 – 2010
Amirkabir University of Technology
GPA: 18.86/20

Selected Publications

[1] **Gholamali Aminian**, Idan Shenfeld, Amir R. Asadi, Ahmad Beirami, and Youssef Mroueh. "Best-of-n through the smoothing lens: KL Divergence and Regret Analysis." (ICLR 2026).

[2] **Gholamali Aminian**, Amir R. Asadi, Idan Shenfeld, and Youssef Mroueh. "KL-Regularized RLHF with Multiple Reference Models: Exact Solutions and Sample Complexity." (NeurIPS 2025)

[3] **Gholamali Aminian**, Amir R.Asadi, Tian Li, Ahmad Beirami, Gesine Reinert, Samuel N. Cohen. "Generalization and Robustness of the Tilted Empirical Risk". (ICML 2025)

[4] Armin Behnamnia*, **Gholamali Aminian***, Alireza Aghaei, Chengchun Shi, Vincent Y. F. Tan, Hamid R. Rabiee, "Log-Sum-Exponential Estimator for Off-Policy Evaluation and Learning." (ICML 2025, Spotlight)

[5] **Gholamali Aminian***, Yuheng Bu*, Laura Toni, Miguel Rodrigues, and Gregory Wornell. "An Exact Characterization of the Generalization Error for the Gibbs Algorithm." Advances in Neural Information Processing Systems 34 (NeurIPS 2021).

Honors and Awards

- Awarded the **International Newton Fellowship** by Royal Society, 2019
- Ranked **1st** among 90 B.Sc. students in Dept. of Electrical Engineering
- **3rd** rank in National Mathematics Olympiad of Iran, 2005 (Bronze Medal)
- **Best PhD thesis award** from theoretical perspective in EE department of Sharif University
- **Iran National Elite Foundation Fellowship** in B.Sc. and M.Sc. periods

Teaching and Mentorship

Mentoring Graduate Students:

- Lorcan O'Connor, M.Sc. candidate at Oxford University — Summer 2023
- Zhaoyan Lyu, Ph.D. candidate at UCL EE — Summer 2020 – Fall 2022
- Haiyun He, Ph.D. candidate at NUS — Fall 2021 – Fall 2022

Teaching Experience:

- Information Theory, Oxford — Fall 2023-2024
- ELEC0134 - Applied Machine Learning Systems, UCL — Fall 2020-2021
- ELEC0139 - Emerging Topics in ML Systems, UCL — Spring 2020-2021
- Information Theory, Sharif University — Fall 2016-2017

Research Interests

- **Agentic AI:** evaluation, reliability and quality control of LLM agents; inference-time algorithms (best-of-n, verifiers)
- **LLM Post-training and mid-training** (KL-regularized RLHF, reward modelling)
- **Reinforcement learning** (off-policy evaluation and learning)
- **Theoretical Inspired Machine Learning Algorithms**
  - Theoretical foundation of Trustworthy AI
  - Semi-supervised learning
  - Theoretical inspired algorithms for Transfer learning and Meta-Learning
- **Optimization Theory**

Technical Skills

- **Programming Languages:** Python (Advanced), C++ (Intermediate), Java (Intermediate)
- **Scientific Computing:** MATLAB (Advanced), Simulink
- **Document Preparation:** LaTeX (Advanced)
- **Machine Learning:** PyTorch, TensorFlow, Scikit-learn
- **LLM & Agents:** Hugging Face Transformers, LLM evaluation frameworks, agent orchestration and tool-use, prompt/rubric design

Professional Activities

- **Professional Memberships:** IEEE Information Theory Society
- **Journal Reviewer:** IEEE Transactions on Information Theory, JMLR, TMLR, Entropy
- **Conference Reviewer:** AISTATS, NeurIPS, ICLR, ICML, ISIT, ITW

Invited Talks

- "Information-theoretical Analysis of KL-regularized RLHF with Multiple Reference Models"

  Location: Department of Pure Mathematics and Mathematical Statistics at the University of Cambridge

  Invited talk at The Eighth London Symposium on Information Theory
- "Symmetrized KL Information: Channel Capacity and Learning Theory"

  Location: Department of Pure Mathematics and Mathematical Statistics at the University of Cambridge

  Host: Prof. Ramji Venkataramanan
- "Mean-field Analysis of Generalization Errors"

  Location: Department of Mathematical Sciences, Liverpool University

  Host: Dr. Ehsan Azmoodeh,

- "Mean-field Analysis of Generalization Errors"

  Location: Mathematical and Computational Finance Seminars at Mathematical Institute (University of Oxford, Radcliffe Observatory, Andrew Wiles Building),

  Host: Prof. Rama Cont,

- "Mean-field Analysis of Generalization Errors"

  Location: Équipe NEO, Centre Inria d'Université Côte d'Azur INRIA, Sophia Antipolis, France.

  Host: Dr. Samir M. Perlaza,

- "Generalization Error of Graph Neural Networks in the Mean-field Regime"

  Location: Statistic Department, Oxford University, UK,

  Host: Prof. Gesine Reinert,

Publication List

Preprints:

[1]. **Gholamali Aminian**, Samuel N. Cohen, and Łukasz Szpruch. "Mean-field Analysis of Generalization Errors."

[2]. **Gholamali Aminian**, Łukasz Szpruch, and Samuel N. Cohen. "Understanding Transfer Learning via Mean-field Analysis" .

Conference Papers:

[3]. Carla Wirth, Leoni, **Gholamali Aminian**, and Gesine Reinert. "Private Synthetic Graph Generation and Fused Gromov-Wasserstein Distance." (AISTATS 2026)

[4]. **Gholamali Aminian**, Idan Shenfeld, Amir R. Asadi, Ahmad Beirami, and Youssef Mroueh. "Best-of-n through the smoothing lens: Kl divergence and regret analysis." (ICLR 2026).

[5]. **Gholamali Aminian**, Andrew Elliott, Tiger Li, Timothy Cheuk Hin Wong, Victor Claude Dehon, Lukasz Szpruch, Carsten Maple, Christopher Read, Martin Brown, Gesine Reinert, Mo Mamouei. "FraudTransformer: Time-Aware GPT for Transaction Fraud Detection." Conference on AI (CAI 2026)

[6]. **Gholamali Aminian**, Amir R. Asadi, Idan Shenfeld, and Youssef Mroueh. "KL-Regularized RLHF with Multiple Reference Models: Exact Solutions and Sample Complexity." (NeurIPS 2025)

[7]. Xiangkun Wu, Ting Li, **Gholamali Aminian**, Armin Behnamnia, Hamid R Rabiee, Chengchun Shi. "Pessimistic data integration for policy evaluation" (NeurIPS 2025)

[8]. Armin Behnamnia*, **Gholamali Aminian***, Alireza Aghaei, Chengchun Shi, Vincent Y. F. Tan, Hamid R. Rabiee, "Log-Sum-Exponential Estimator for Off-Policy Evaluation and Learning." (ICML 2025, Spotlight)

[9]. **Gholamali Aminian**, Amir R.Asadi, Tian Li, Ahmad Beirami, Gesine Reinert, Samuel N. Cohen. "Generalization and Robustness of the Tilted Empirical Risk". (ICML 2025)

[10]. **Gholamali Aminian***, Yixuan He*, Gesine Reinert, Lukasz Szpruch, Samuel Cohen. "Generalization Error of Graph Neural Networks in the Mean-field Regime." (ICML 2024)

[11]. **Gholamali Aminian,** Amirhossien Bagheri, Mahyar JafariNodeh, Radmehr Karimian, Mohammad-Hossein Yassaee "Robust Semi-supervised Learning via f-Divergence and α-Rényi Divergence" (ISIT 2024)

[12]. **Gholamali Aminian**. " Generalization Bounds Under Heavy-Tailed Losses."

[13]. He, Haiyun, **Gholamali Aminian,** Yuheng Bu, Miguel Rodrigues, and Vincent YF Tan. "How Does Pseudo-Labeling Affect the Generalization Error of the Semi-Supervised Gibbs Algorithm?." International Conference on Artificial Intelligence and Statistics 26 (AISTATS 2023).

[14]. Bu, Yuheng, Harsha Vardhan Tetali, **Gholamali Aminian**, Miguel Rodrigues, and Gregory Wornell. "On the Generalization Error of Meta Learning for the Gibbs Algorithm." In IEEE International Symposium on Information Theory (ISIT),(2023).

[15]. **Gholamali Aminian*,** Roberto Vega*, Omar Rivasplata, Laura toni, Miguel Rodrigues. " Semi-Counterfactual Risk Minimization Via Neural Networks" European Workshop on Reinforcement Learning (EWRL 2022).

[16]. **Gholamali Aminian*,** Yuheng Bu*, Miguel Rodrigues, and Gregory Wornell. "Tighter Expected Generalization Error Bounds via Convexity of Information Measures," In IEEE International Symposium on Information Theory (ISIT),(2022).

[17]. **Gholamali Aminian*,** Mahed Abroshan*, Mahdi Khalili, Laura Toni, Miguel R. D. Rodrigues. "A Theoretical-Inspired Semi-supervised Learning Algorithm under Covariate-shift" International Conference on Artificial Intelligence and Statistics 25 (AISTATS 2022)

[18]. **Gholamali Aminian*,** Yuheng Bu*, Laura Toni, Gregory Wornel, Miguel R. D. Rodrigues, "Characterizing and Understanding the Generalization Error of Transfer Learning with Gibbs Algorithm" International Conference on Artificial Intelligence and Statistics 25 (AISTATS 2022)

[19]. **Gholamali Aminian*,** Yuheng Bu*, Laura Toni, Miguel Rodrigues, and Gregory Wornell. "An Exact Characterization of the Generalization Error for the Gibbs Algorithm." Advances in Neural Information Processing Systems 34 (NeurIPS 2021).

[20]. Lyu, Zhaoyan, **Gholamali Aminian,** and Miguel RD Rodrigues. "Toward Minimal-Sufficiency in Regression Tasks: An Approach Based on a Variational Estimation Bottleneck." In 2021 IEEE 31st International Workshop on Machine Learning for Signal Processing (MLSP), pp. 1-6. IEEE, 2021.

[21]. **Gholamali Aminian,** Laura Toni, and Miguel RD Rodrigues. "Information-Theoretic Bounds on the Moments of the Generalization Error of Learning Algorithms." In IEEE International Symposium on Information Theory (ISIT),(2021).

[22]. **Gholamali Aminian,** Laura Toni, and Miguel RD Rodrigues. "Jensen-Shannon Information Based Characterization of the Generalization Error of Learning Algorithms." Information Theory Workshop (ITW) (2020).

[23]. **Gholamali Aminian,** Mahtab Mirmohseni, Masoumeh Nasiri Kenari, and Faramarz Fekri. "On the capacity of level and type modulations in Molecular communication with ligand receptors." In IEEE International Symposium on Information Theory (ISIT), 2015 , pp. 1951-1955. IEEE, 2015.

[24]. **Gholamali Aminian,** Hamidreza Arjmandi, Amin Gohari, Masoumeh Nasiri Kenari, and Urbashi Mitra. "Capacity of LTI-Poisson channel for diffusion based molecular communication." In Communications (ICC), 2015 IEEE International Conference on, pp. 1060-1065. IEEE, 2015.

[25]. **Gholamali Aminian,** Hamid Ghourchian, Amin Gohari, Mahtab Mirmohseni, and Masoumeh Nasiri-Kenari. "On the capacity of signal dependent noise channels." In Communication and Information Theory (IWCIT), 2017 Iran Workshop on, pp. 1-6. IEEE, 2017.

[26]. Maryam Farahnak Ghazani, **Gholamali Aminian,** Mahtab Mirmohseni, Amin Gohari, and Masoumeh Nasiri Kenari. "Physical layer network coding in molecular two-way relay networks." In Communication and Information Theory (IWCIT), 2016 Iran Workshop on, pp. 1-6. IEEE, 2016.

Journal Papers:

[27]. **Gholamali Aminian*,** Yuheng Bu*, Laura Toni, Miguel Rodrigues, and Gregory Wornell. " Information-theoretic Characterizations of Generalization Error for the Gibbs Algorithm," IEEE Transactions on Information Theory, Oct. 2023.

[28]. Lyu, Zhaoyan, **Gholamali Aminian**, and Miguel RD Rodrigues. "On Neural Networks Fitting, Compression, and Generalization Behavior via Information-Bottleneck-like Approaches." Entropy 25.7 (2023): 1063.

[29]. **Gholamali Aminian*,** Saeed Masiha*, Laura Toni, and Miguel Rodrigues "Characterization of the Upper Bound on Generalization Error of Learning Algorithms Using Auxiliary Distribution" IEEE Journal on Selected Areas in Information Theory, 2024.

[30]. **Gholamali Aminian,** Hamidreza Arjmandi, Amin Gohari, Masoumeh Nasiri Kenari, and Urbashi Mitra. "Capacity of Diffusion-based Molecular Communication Networks over LTI-Poisson Channels." IEEE Transactions on Molecular, Biological and Multi-Scale Communications, Volume:1 , Issue: 2, pp. 188 - 20. 2016.

[31]. **Gholamali Aminian,** Maryam Farahnak Ghazani, Mahtab Mirmohseni, Masoumeh Nasiri Kenari, and Faramarz Fekri. "On the Capacity of Point-to-Point and Multiple-Access Molecular Communications with Ligand-Receptors." IEEE Transactions on Molecular, Biological and Multi-Scale Communications, Volume:1 , Issue: 4, pp. 331-346. 2016.

[32]. Hamid Ghourchian, **Gholamali Aminian,** Amin Gohari, Mahtab Mirmohseni, and Masoumeh Nasiri-Kenari. "On the Capacity of a Class of Signal-Dependent Noise Channels." IEEE Transactions on Information Theory 64, no. 12 (2018): 7828-7846.

[33]. Hamid G. Bafghi, Amin Gohari, Mahtab Mirmohseni, **Gholamali Aminian,** and Masoumeh Nasiri-Kenari. "Diffusion-based molecular communication with limited molecule production rate." IEEE Transactions on Molecular, Biological and Multi-Scale Communications 4, no. 2 (2018): 61-72.

[34]. Maryam Farahnak-Ghazani, **Gholamali Aminian,** Mahtab Mirmohseni, Amin Gohari, and Masoumeh Nasiri-Kenari. "On medium chemical reaction in diffusion-based molecular communication: A two-way relaying example." IEEE Transactions on Communications 67, no. 2 (2018): 1117-1132.

Workshop Papers:

[35]. **Gholamali Aminian,** Inhee Park, Xiaoting Kuang, Ghazal Azarfar, Ehsan Karimiara, David Jeremiah, Sisodiya Dilawar Singh, Mridul Garg, Chao Han, Sindhur U. Joshi, Tatiany Soratto, Saurabh Verma, Sergio Marinho da Silva, Aayushi Arya, Viren Pattni, Saurabh G. Choudhary, Abbas Hosseini. "ML4Science-Bench: Benchmarking Autonomous Agents on End-to-End Machine Learning for Science" NeurIPS 2026 Workshop · AI for Science · **Oral**

[36]. Bagheri, Amirhossein, Radmehr Karimian, and **Gholamali Aminian**. "f-SCRUB: Unbounded Machine Unlearning Via f-divergences." ICLR 2025 Workshop on Navigating and Addressing Data Problems for Foundation Models.

[37]. Armin Behnamnia*, **Gholamali Aminian***, Chengchun Shi, Vincent Tan, Hamid R. Rabiee." Batch Learning via Log-Sum-Exponential Estimator from Logged Bandit Feedback" ICML 2024 Workshop: Aligning Reinforcement Learning Experimentalists and Theorists.

[38]. **Gholamali Aminian**, Idan Shenfeld, Amir R. Asadi, Ahmad Beirami and Youssef Mroueh. "Best-of-N through the Smoothing Lens: KL Divergence and Regret Analysis" ICML 2025 Workshop ES-FoMo III

[39]. **Gholamali Aminian***, Yuheng Bu*, Laura Toni, Miguel R. D. Rodrigues, Gregory Wornell "Characterizing the Generalization Error of Gibbs Algorithm with Symmetrized KL information" ICML-21 Workshop on Information-Theoretic Methods for Rigorous, Responsible, and Reliable Machine Learning

[40]. Zhaoyan Lyu, Gholamali Aminian, Miguel RD Rodrigues. "Synthetic Labeling: A Novel Approach to Advancing Few-Shot Learning" ICLR Tiny-paper